Gayatri Vidya Parishad College of Engineering Autonomous

Electrical and Electronics Engineering | Course Code: EE201

Electrical Circuit Analysis - Laboratory Manual & Practical Guide

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Syllabus Focus: EE201 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Network Theorems DC & AC

" Syllabus Scope: Thevenin, Norton, Superposition, Maximum Power Transfer, Reciprocity, and Millman theor

Unit 2: Transient Response of First-Order Circuits

" Syllabus Scope: Natural and forced response of RL and RC circuits, Differential equations, Time constan

EE201: Electrical Circuit Analysis - Advanced Topics & Assessment

Academic Track: Electrical and Electronics Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Transient Response of Second-Order RLC Circuits

" Series and parallel RLC transient response, Overdamped, critically damped, and underdamped response regimes,

Unit 4: Laplace Transform in Circuit Analysis

" Transformed circuit impedance models of R, L, and C, Initial and final value theorems, Complex frequency s-p

Unit 5: Two-Port Network Parameters

" Impedance Z, Admittance Y, Hybrid H, and Transmission ABCD parameters, Interconnection of two-port networks

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Experimental Verification of Thevenin, Norton, and Superposition Theorems on Resistive DC Networks

Lab 2: Verification of Maximum Power Transfer Theorem for DC and AC Circuits with Complex Load Matching

Lab 3: Determination of Transient Time Constants for First-Order RL and RC Circuits on Digital Storage Os

Lab 4: Experimental Determination of Two-Port Z and Y Parameters of a Symmetrical T and Pi Network

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* William H. Hayt, Jack E. Kemmerly - Engineering Circuit Analysis McGraw-Hill 8th Ed

* Charles K. Alexander, Matthew N.O. Sadiku - Fundamentals of Electric Circuits McGraw-Hill

Authorized NoteVault Educational Document " Verified for Registered University Students